

STARTUP VALIDATION REPORT

Circuit.IQ

AI-Powered Virtual Physics Laboratory for Indian Schools, Colleges & Engineering Institutes

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1. Executive Summary

Circuit.IQ is an AI-powered virtual physics laboratory that lets students run interactive digital experiments — building circuits, adjusting variables, and observing real-time simulated outcomes — instead of only reading theory. It targets a well-documented and structural gap in Indian science education: most schools, colleges, and coaching institutes lack sufficient physical lab infrastructure, and existing online platforms are theory-heavy with limited hands-on interactivity.

The founder has already shipped a working MVP capable of handling 100+ concurrent simulations at sub-second latency, demonstrating real technical execution ability. What is missing at this stage is **market validation**: no structured customer interviews, no pilot schools, and no paying customers yet. This report treats Circuit.IQ as pre-revenue, pre-validation, and evaluates it on market size, competitive positioning, customer fit, and risk — before recommending a concrete path forward.

6.5/10Founder-Market
Fit Score**\$150M**Estimated
TAM (India)**\$22M**Estimated
SAM (3-yr reach)**CONDITION
AL
GO**Overall
Recommendation

Key Findings

- The problem is **real and structurally validated** by government-scale precedent — Amrita/OLabs alone has reached roughly 8,000 schools and ~4 lakh students with free virtual lab content, confirming latent institutional demand for exactly this category.
- India's EdTech market is large (\$3.63B in 2025, growing at ~28% CAGR) but the specific interactive-physics-simulation niche is a small, under-served slice dominated by either free government tools (OLabs, PhET) or expensive global platforms (Labster) not localized for Indian curricula.
- Circuit.IQ's core differentiation opportunity is the white space between "free but generic" and "powerful but expensive/foreign": an AI-personalized, India-curriculum-aligned, affordably-priced virtual lab.
- The single biggest gap today is **go-to-market**, not product: zero structured interviews, no pilot institution, and no revenue signal. Institutional sales cycles in Indian education (school/college procurement) are also inherently slow (2–6 months), which matters for a solo-founder side project.
- Recommendation: treat the next 30–60 days as a **validation sprint** (structured interviews + one free pilot), not a scaling sprint. A Go decision on scaling should be conditional on securing at least one paying or committed pilot institution.

2. Problem Validation

Is this a real, painful, and widespread problem?

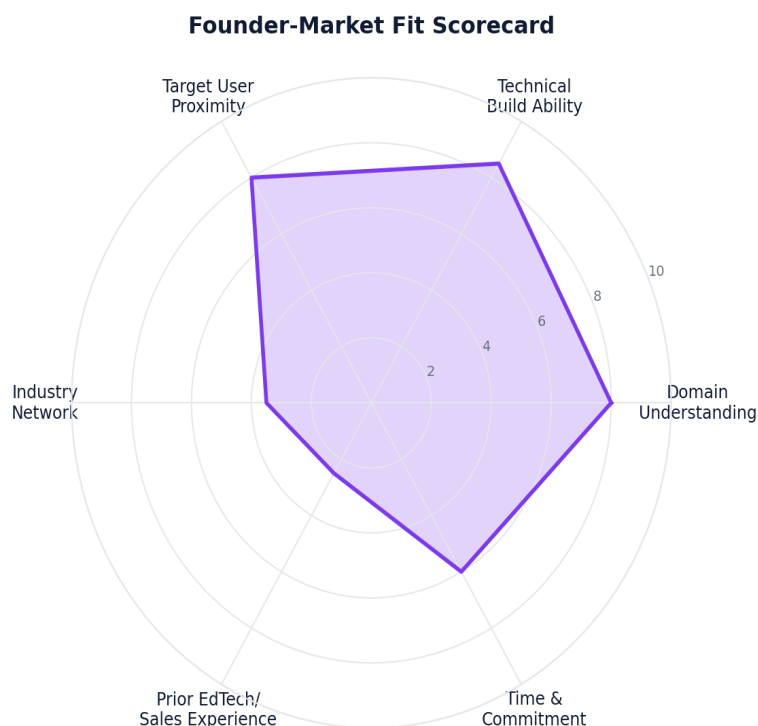
Yes — with strong third-party corroboration. India has approximately 1.5 million schools and 52,321 colleges, the vast majority of which are still in the early stages of structured digital lab adoption. Government-funded initiatives such as OLabs (Amrita Vishwa Vidyapeetham, with CDAC and the Ministry of Education/MeitY) exist specifically because physical lab access is inconsistent, equipment is costly or scarce, and lab time is constrained — the exact problem statement in the founder's pitch. OLabs alone has been adopted by roughly 8,000 schools and trained about 28,000 teachers, which is strong evidence the underlying pain point is real at national scale.

Evidence Table

Evidence Type	Signal	Strength
Government-scale precedent	OLabs: ~8,000 schools, ~4 lakh students, ~28,000 teachers trained on free virtual labs (Physics/Chem/Bio, Classes 9–12)	Strong
Global commercial precedent	Labster (Denmark/US) monetizes virtual STEM labs institutionally at \$79–109/student/year across universities & high schools worldwide	Strong
Category-level market growth	India EdTech market \$3.63B (2025) → \$33.31B (2034 est.), 27.9% CAGR; K-12 is the largest single segment at 43% share	Moderate (broad category, not lab-specific)
Direct customer validation	No structured interviews conducted; no LOIs, pilots, or paying users yet	Weak — not yet done

*Verdict: The **category-level problem** is well validated by precedent (government adoption + global commercial traction). The **founder-specific** problem-solution fit for Circuit.IQ specifically is not yet validated — this is the single most urgent gap to close before further building.*

3. Founder-Market Fit Score



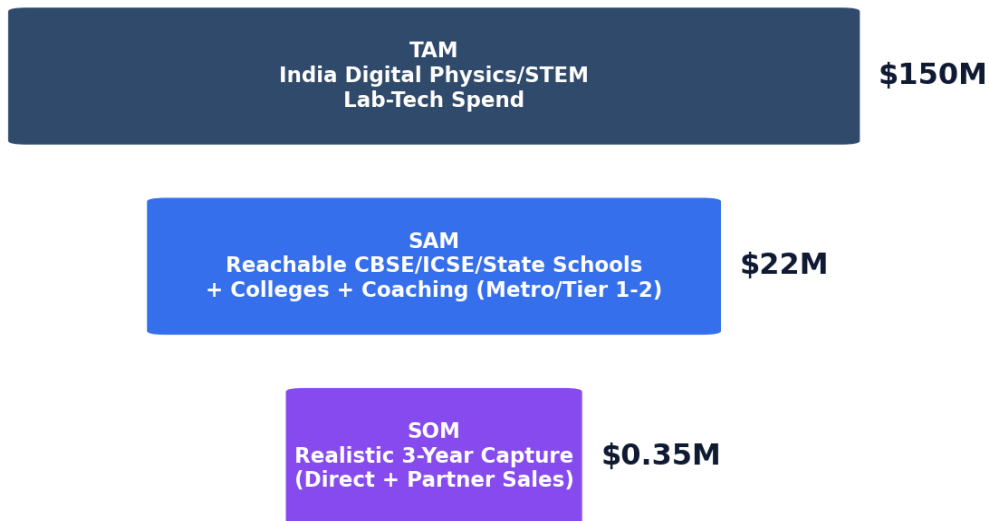
Dimension	Score /10	Rationale
Domain understanding	8.0	Engineering student who has personally experienced lab-access constraints; understands the pedagogy gap first-hand
Technical build ability	8.5	Already shipped a working MVP (React/TypeScript/Python/FastAPI) handling 100+ concurrent simulations at sub-second latency
Target user proximity	8.0	Is literally inside the target demographic (engineering/college student segment); easy access to peer feedback
Industry network	3.5	No disclosed relationships with school administrators, principals, college HODs, or EdTech distribution partners yet
Prior EdTech/sales experience	2.5	No prior track record of institutional sales, B2B/B2G EdTech deals, or structured customer discovery
Time & commitment	6.0	Currently a side project alongside a B.Tech degree and job search — real but split attention

Composite Founder-Market Fit Score: 6.1 / 10 — strong on product execution and domain empathy, weak on distribution and commercial experience. This is a common and fixable profile for a technical solo founder; it is not a disqualifying weakness, but it should directly shape the 30-day plan (Section 15) toward customer conversations rather than more building.

4. TAM · SAM · SOM Analysis

Circuit.IQ sits inside India's broader EdTech market, valued at approximately \$3.63B in 2025 and forecast to reach \$33.31B by 2034 at a 27.9% CAGR, with K-12 as the largest single sub-segment (43% share in 2025). Digital lab/practical-simulation tools are a narrow slice of this — most reported market-size figures for Indian EdTech blend tutoring, test-prep, and content platforms, none of which isolate "virtual lab" spend. The estimates below are therefore built **bottom-up** from institution counts rather than top-down from blended industry figures, and should be read as directional, not audited.

TAM → SAM → SOM (Approx. USD, India Market)



Layer	Definition	Sizing Logic	Est. Value
TAM	All India spend on digital physics/STEM lab simulation tools across schools, colleges, coaching institutes and EdTech platforms	~1.5M schools (subset with science stream) + 52,321 colleges + coaching institutes; blended addressable license/subscription spend	~\$150M / yr
SAM	Reachable segment: private CBSE/ICSE/state-board schools, engineering colleges, and coaching institutes in metro + tier 1-2 cities with existing digital budgets, reachable via direct + partner sales in 3 years	~75,000 institutions × ~\$25,000 (~\$300) average annual license value	~\$22M / yr
SOM	Realistic capture in Year 1–3 for a solo-founder-led side project scaling to a small team, focused on schools + engineering colleges in J&K, Delhi NCR, and 1-2 metro clusters	750–1,500 institutions by Year 3 at ~\$25,000 avg. contract value	~\$225K–\$450K / yr by Yr 3

*Reality check: SOM assumes disciplined institutional sales execution most solo student-founders under-resource. A more conservative Year-1 target is **10–30 pilot institutions** (schools + one engineering college), generating proof points rather than material revenue — revenue scale-up is a Year 2–3 story, not a Year 1 one.*

5. Competitor Analysis

Competitor	Type	Physics Coverage	Price	AI Personalization	India Curriculum Fit	Key Weakness
OLabs (Amrita/CDAC, Govt. of India)	Free, govt-backed	Classes 9–12, NCERT/SCERT aligned	Free	None	Very High	Static content, no AI adaptivity, no engineering-level depth, UX dated
PhET (Univ. of Colorado)	Free, global open-source	Broad, university-grade sims	Free	None	Low (not India-curriculum mapped)	Generic 2D sims, not gamified, no local language or board alignment
Labster	Global commercial (premium)	Physics + Chem + Bio, immersive 3D	\$79–109/student/yr	Limited adaptivity	Low	Expensive for Indian budgets, not localized, heavier for K-12 use
Vedantu / PhysicsWallah / BYJU'S (broad EdTech)	Commercial, India-focused	Video/theory-first, limited interactivity	■varies, subscription	Some AI tutoring features	High	Practical/lab interactivity is a minor feature, not the core product
Circuit.IQ (this venture)	Early-stage, India-focused	Physics-specific, engineering-adjacent depth	TBD (target: affordable SaaS)	AI-personalized (core differentiator)	High (if built intentionally)	No brand, no distribution, no institutional proof points yet

The clearest strategic read: competitors cluster into two camps — **free-but-generic** (OLabs, PhET) and **powerful-but-expensive/foreign** (Labster). Broad EdTech players (PhysicsWallah, Vedantu, BYJU'S) treat lab interactivity as a minor feature bolted onto video-first products. This leaves a genuine white space for an AI-personalized, India-curriculum-native, affordably-priced virtual physics lab — provided Circuit.IQ can prove measurable learning outcomes and win institutional trust without an established brand.

6. Market Gap Analysis

Gap	Why It Exists	Circuit.IQ's Opportunity
AI-personalized feedback in physics simulations	Free tools (OLabs, PhET) are static content; premium tools (Labster) personalize course structure but not per-student remediation	Use AI to detect misconceptions in real time and adapt difficulty/hints per student
Affordable engineering-level depth	Labster's depth comes at global pricing; Indian engineering colleges/coaching institutes are price-sensitive	Position between free (too basic) and Labster (too expensive) with India-priced SaaS tiers
Curriculum-native content for both school AND engineering syllabi	Most tools serve either K-12 (OLabs, PhET) or higher-ed/global (Labster) — few span both CBSE/ICSE school physics and first/second-year engineering physics	Single platform spanning Class 9–12 and B.Tech first-year physics, single sales motion into schools + colleges
Teacher-facing analytics for lab performance	Government tools have limited dashboards; commercial tools' dashboards are built for global LMS integration, not Indian school workflows	Build lightweight, India-context teacher dashboards (WhatsApp/Excel-export friendly, low-bandwidth)

7. Ideal Customer Profile (ICP)

Attribute	Profile
Institution type	Private CBSE/ICSE schools (Classes 9–12) and Tier 2/3 engineering colleges/polytechnics
Geography (Phase 1)	Jammu & Kashmir, Delhi NCR, and 1–2 additional metro clusters (Bangalore/Pune/Hyderabad) with digital-adoption momentum
Size	300–2,000 students; mid-sized institutions that feel lab-infrastructure cost pressure but have some digital budget
Budget authority	Principal/Director (schools) or HOD/Dean Academics (colleges), often with input from an IT/digital-learning coordinator
Trigger conditions	Upcoming NEP 2020 compliance push, lab equipment shortage/damage, low practical exam scores, push for "smart classroom" branding
Buying behavior	Prefers pilot-then-scale; wants free trial period and visible learning-outcome proof before annual commitment
Disqualifiers	Institutions with zero digital budget, no computer lab/device access, or fully offline rural schools with no internet connectivity (Phase 1 exclusion, not permanent)

8. Buyer Personas

Persona 1 — Ritu Sharma, School Principal (Private CBSE School, Tier-2 city)

"I want my students to score well in practicals and board exams, but I can't justify a new lab budget every year."

Goals	Improve practical exam scores; showcase "smart lab" credentials to parents for admissions; comply with NEP 2020 digital-learning mandates.
Frustrations	Limited annual budget cycle (approved once a year); skeptical of unproven EdTech vendors after past disappointments with subscriptions that were never used by teachers; low tolerance for complex onboarding.
Decision Criteria	Needs a free/low-cost pilot term, clear before/after learning outcome data, and minimal teacher training overhead.

Persona 2 — Dr. Anil Kumar, HOD Physics / Dean Academics (Engineering College)

"Our first-year physics lab equipment is shared across 400 students and half of it is outdated — students barely get hands-on time."

Goals	Reduce lab equipment strain and scheduling bottlenecks; improve first-year retention in core physics; satisfy AICTE/university accreditation on practical exposure.
Frustrations	Procurement requires committee approval; wary of tools that don't map cleanly to the university's prescribed lab manual/syllabus.
Decision Criteria	Needs explicit mapping to the university's official lab manual, multi-user concurrent access, and usage analytics for accreditation reporting.

Persona 3 — Aditya, Class 11 Science Student (Self-directed learner)

"I understand physics formulas but I've never actually built a circuit — I want to see what happens, not just memorize it."

Goals	Score well in board/JEE practical components; build genuine conceptual intuition, not rote memorization; study at odd hours outside school lab timings.
Frustrations	Free/cheap alternatives already exist (YouTube, PhET); low willingness to pay individually unless bundled through school or coaching institute.
Decision Criteria	Needs a smooth, mobile-friendly, low-data experience; more likely to be a B2B2C user via school/coaching license than a direct payer.

9. Customer Pain Points

Pain Point	Segment	Severity
Physical lab equipment is scarce, shared, or damaged, limiting hands-on time per student	Schools & Colleges	High
Existing online content (videos, PDFs, LMS) is theory-heavy with little true interactivity	All segments	High
Free tools (OLabs/PhET) lack adaptivity, gamification, and modern UX that hold student attention	Students	Medium
No easy way for teachers to track which specific concepts students are struggling with in practicals	Teachers/HODs	Medium
Budget cycles and procurement bureaucracy make institutions risk-averse toward new EdTech vendors	Principals/Deans	High
Rural/low-bandwidth environments make heavier 3D platforms (like Labster) impractical	Schools (Tier 2/3)	Medium

10. Buying Triggers & Objections

Buying Triggers

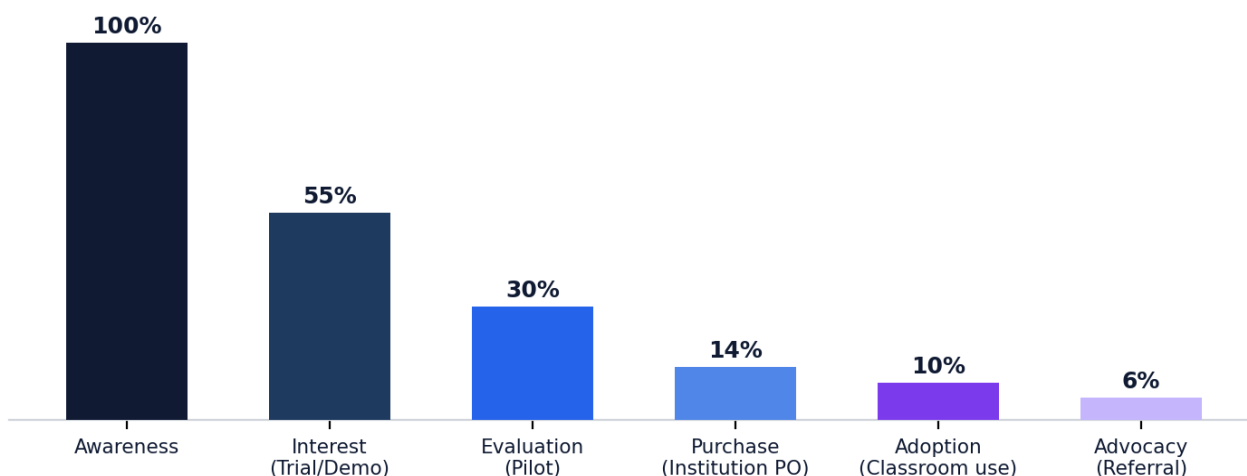
- NEP 2020 compliance pressure pushing schools toward demonstrable "digital/practical learning" initiatives.
- A poor batch of practical exam results or accreditation feedback citing weak lab exposure.
- Lab equipment damage/shortage creating an immediate scheduling crunch.
- Competitive pressure — a nearby school/college publicly adopting a "smart lab" or AI-learning tool.
- Start of a new academic year or admission cycle, when marketing-facing upgrades are budgeted.

Common Objections & Responses

Objection	Suggested Response Strategy
"We already use O Labs/PhET for free — why pay?"	Position as a complement, not replacement: AI-personalized feedback, engagement/gamification, and progress analytics that free tools don't offer.
"We don't have budget approved this cycle."	Offer a no-cost pilot term (1 term/semester) tied to a specific measurable outcome (e.g., practical score improvement) to justify next cycle's budget.
"How do we know this actually improves learning outcomes?"	Run and publish a small pilot case study with before/after data — this is the single highest-leverage asset to build in the next 60–90 days.
"Our internet/device infrastructure is limited."	Ensure a lightweight, low-bandwidth mode; emphasize this explicitly in early sales conversations with Tier 2/3 institutions.
"You're a student founder with no institutional track record."	Lean into transparency — frame as a founder deeply embedded in the user problem; use pilot results and testimonials to substitute for brand credibility.

11. Customer Journey

Illustrative Institutional Buyer Journey — Conversion Funnel



Stage	What Happens	Circuit.IQ's Job-to-be-Done
Awareness	Principal/HOD hears about Circuit.IQ via LinkedIn content, referral, or direct outreach	Publish credible, education-focused content (already underway via ABTalks series) and targeted outreach to school/college networks
Interest (Trial/Demo)	Decision-maker requests a demo or trial account	Provide a frictionless, fast demo — ideally a live link, no install, works on low-end devices
Evaluation (Pilot)	Institution runs a limited pilot with 1–2 classes over 4–8 weeks	Support pilot actively; collect usage + outcome data; check in with teachers weekly
Purchase (Institution PO)	Formal approval/purchase order or subscription signed, typically after budget-cycle alignment	Provide simple, low-friction pricing/invoicing suited to Indian institutional procurement (no complex enterprise contracts)
Adoption (Classroom use)	Teachers integrate it into regular lesson plans	Ongoing lightweight teacher support/training; usage nudges
Advocacy (Referral)	Satisfied principal/HOD refers peer institutions	Systematically request referrals/testimonials/case studies from every successful pilot

12. Risk Assessment

Risk Assessment Matrix



Risk	Likelihood	Impact	Mitigation
Long institutional sales cycles stall momentum	High	High	Run parallel pilots with multiple institutions simultaneously; avoid single-customer dependency
Free government alternative (OLabs) undercuts willingness to pay	High	High	Differentiate clearly on AI personalization, analytics, and UX rather than competing on content breadth alone
No dedicated sales/BD resource (solo founder, side project)	Medium-High	Medium-High	Time-box a structured outreach sprint (Section 15); consider a co-founder/intern for BD once first pilot lands
Content breadth gap vs. Labster/PhET at scale	Medium	Medium	Focus initial content depth narrowly (Class 11-12 + first-year engineering physics) rather than trying to match breadth immediately
Device/bandwidth constraints in target schools	Medium	Medium	Build and test a genuinely low-bandwidth mode; validate on actual school devices, not just personal laptops
Founder solo/bandwidth risk (balancing degree + job search + venture)	Medium	Medium	Treat this explicitly as a side project until a pilot validates demand; avoid over-investing before validation
Data privacy / child-data compliance (minors using the platform)	Low-Medium	Medium-High	Adopt minimal data collection by default; avoid storing unnecessary student PII; review India's DPDP Act obligations before scaling
Curriculum drift vs. board syllabus changes	Low	Low-Medium	Build a lightweight content-update process tied to CBSE/state board syllabus revision cycles

13. Pivot Opportunities

Pivot Option	Description	When to Consider
B2B2C via coaching institutes	Sell through JEE/NEET coaching chains (Allen, Aakash-adjacent regional players) rather than schools directly — faster sales cycles, existing digital-tool budgets	If school procurement proves too slow after 60–90 days of outreach
White-label / API licensing to larger EdTech platforms	License the simulation engine to existing platforms (Vedantu, PhysicsWallah-type players, LMS providers) instead of selling direct to institutions	If direct institutional sales prove structurally too slow for a solo founder
Expand beyond physics to full STEM virtual lab suite	Broaden to chemistry/biology simulations once physics engine and sales motion are proven, competing more directly with O Labs/Labster breadth	After achieving repeatable sales in physics-only vertical
Shift to individual/student-direct freemium model	Bypass institutional sales entirely; monetize students directly via freemium + premium AI-tutoring add-ons	If institutional sales cycles remain a persistent bottleneck and self-serve traction (from content/LinkedIn audience) is strong
Corporate/skilling adjacency	Extend simulation approach to technical upskilling (electronics, embedded systems) for the broader AI/EdTech skilling market	Long-term option once core India-school/college motion is proven and founder wants larger TAM

14. Go / No-Go Recommendation

CONDITIONAL GO — Validate Before You Scale

Circuit.IQ has real technical execution behind it and targets a structurally validated problem — but it has **zero direct customer validation** so far. The recommendation is not "stop" and not "scale immediately"; it is to spend the next 30–60 days converting the existing MVP into structured proof — via customer interviews and at least one committed pilot institution — before investing further founder time, and certainly before any paid acquisition spend or hiring.

Conditions for upgrading to a full GO

- At least 8–12 structured interviews completed with principals, HODs, or teachers (not just students) confirming willingness to adopt.
- One school or college agrees to a free/low-cost pilot for a full term/semester.
- Pilot produces a measurable outcome signal (engagement, practical scores, or teacher-reported time saved).
- At least one institution indicates verbal or written willingness to pay for continued/expanded access after the pilot.

Conditions that would indicate NO-GO (or pivot)

- Repeated inability to get even one institution to commit to a free pilot after genuine, sustained outreach (60+ days).
- Pilot data shows no measurable engagement or learning benefit over existing free tools (OLabs/PhET).
- Consistent objection that "free is good enough" with no willingness to pay even a nominal fee post-pilot.

15. 30-Day Action Plan

Week	Focus	Key Actions	Output
Week 1 (Days 1–7)	Interview prep & outreach list	Draft a 10-question customer interview script for principals/HODs/teachers; build a target list of 30–40 schools/colleges (start in Srinagar/J&K + existing network); reach out via LinkedIn, email, and personal contacts	Interview script + outreach list of 30–40 target institutions
Week 2 (Days 8–14)	Conduct interviews	Complete 8–12 structured conversations with principals, HODs, or senior science teachers; record objections, willingness to pay, and current tools in use	8–12 completed interviews with synthesized findings
Week 3 (Days 15–21)	Secure one pilot commitment	Approach the 2–3 most promising institutions from interviews with a concrete, low-friction pilot offer (free, 4–8 weeks, 1–2 classes); finalize pilot scope and success metrics	1 signed/verbal pilot commitment with defined success metrics
Week 4 (Days 22–30)	Launch pilot & instrument feedback	Onboard teacher(s) and students on Circuit.IQ; set up lightweight usage tracking and a simple pre/post concept-understanding quiz; schedule a mid-pilot check-in call	Live pilot running with baseline data captured + mid-point review scheduled

Guiding Principle for the Next 30 Days

Do not write more product code this month unless it directly unblocks the pilot. The MVP is already technically credible — the binding constraint right now is customer proof, not features. Every day should move toward either a completed interview, a pilot commitment, or pilot execution.

16. Sources & Methodology

This report combines founder-provided information with publicly available secondary research (market-sizing reports, competitor websites/reviews, and industry press) current as of July 2026. TAM/SAM/SOM, buyer personas, pricing benchmarks, and risk likelihoods are directional estimates built from available public data and standard startup-advisory frameworks — not audited financial or primary-research figures. All figures should be revisited once direct customer interviews (Section 15) are complete.

- IMARC Group — India Edtech Market Size, Share, and Forecast 2026–2034
- India Market Entry — Beyond the Two-Horse Race: The \$29 Billion EdTech Opportunity in India
- MarketsandMarkets — India EdTech Market Size, Share & Growth Forecast to 2030
- Amrita Vishwa Vidyapeetham — OLABS / Virtual Labs project pages (vlab.amrita.edu, olabs.edu.in)
- Amma.org — coverage of OLABS reach (schools, students, teachers trained)
- Labster — product, pricing and review pages (Capterra, G2, EdTech Impact, 360Quadrants)
- PhET Interactive Simulations — University of Colorado Boulder
- RAYSolute — Indian EdTech Market 2026 Analysis